



- **Engineering Ethics**
- **Stakeholder Engagement & Management**



Engineering Ethics

- As a profession, engineers have the public trust and attempt to apply the principles of science and mathematics to develop practical and economical solutions to technical problems
- Work is the link between scientific discoveries and the commercial applications that meet societal and consumer needs.



What does Engineering Ethics bring to your mind ?



Ethical Model

- Virtue Ethics vs Social Contract Theory
- Duty Ethics vs Utilitarianism

Virtue

- Character traits associated with human excellence
- Exercising these traits contributes to one's own or society's well-being
- Examples : generosity, courage, kindness, benevolence

Vice

- Character traits associated with human weakness
- Examples: selfishness, and cowardice, and meanness, and malevolence



Virtue Ethics Defined

- Practice of virtue builds the moral character of society
 - generosity, courage, kindness, benevolence
- In contrast, practice vice undermines the moral character of society
 - selfishness, cowardice, meanness, malevolence
- Practice is the key word - practise makes it habit



Contrasts

- Virtue ethics
 - Pursuing positive behavior and avoiding negative behavior
 - Making good choices and that leads to
 - Becoming a good habit and becomes a part of us
- Social Contract Theory
 - Definition of lower threshold that requires punishment



Social Contract Theory

- Social contract theory is the opposite of virtue ethics
- Virtue Ethics defines the upper threshold
- Social Contract Theory defines the lower threshold

That is at what point should be punished for their bad behaviour

So, we strike a deal



Social Contract Theory (Modern)

- Should some group benefit at the expense of others (J Rawls)
- No one deserves the advantage or disadvantage they are born with
 - These are simply matter of luck
- Everyone deserves the opportunity to develop their innate abilities
 - Examples: Merit based and Need based financial need



Duty Ethics (aka Deontological)

- Focus on duties to
 - Other People
 - God
 - Animals etc
- Defined (Immanuel Kant) in terms of absolutes
 - Universal rules that must be followed by everyone

Something you are required to do in all circumstances where you want to or not

- Some of the duties
 - Always tell the truth.
 - Always keep your promises etc



Utilitarianism (aka Consequentialism)

- In duty ethics, intention (aka will) ruled
- In Utilitarianism ethics , it's outcome ruled
- If in ethical dilemma, choose the act that does the most good (or least) harm for the greatest number of people



IEEE CS/ACM Software Engineering Code of Ethics (short version)

Software engineers shall commit themselves to making the analysis, specification, design, development, testing and maintenance of software a beneficial and respected profession. In accordance with their commitment to the health, safety and welfare of the public, software engineers shall adhere to the following

Eight Principles:

- **Public**: Software engineers shall act consistently with the public interest.
- **Client and Employer**: Software engineers shall act in a manner that is in the best interests of their client and employer, consistent with the public interest.
- **Product**: Software engineers shall ensure that their products and related modifications meet the highest professional standards possible.
- **Judgement**: Software engineers shall maintain integrity and independence in their professional judgment.



Contd.

- **Management**: Software engineering managers and leaders shall subscribe to and promote an ethical approach to the management of software development and maintenance.
- **Profession**: Software engineers shall advance the integrity and reputation of the profession consistent with the public interest.
- **Colleagues**: Software engineers shall be fair to and supportive of their colleagues.
- **Self**: Software engineers shall participate in lifelong learning regarding the practice of their profession and shall promote an ethical approach to the practice of the profession

<https://www.computer.org/education/code-of-ethics>



Case Study - VW Emission Scandal

Cars will sometimes be recalled by their manufacturer for safety or environmental issues. But, what if the manufacturer attempted to avoid a recall because of environmental issues? What could the engineers have done? Also, when cars are recalled, where do they go?



Back Story

How does a Microwave oven works ?

- Magnetron
- Waveguide
- Food sits on turntable
- Microwave bounce off reflective metal walls
- Vibrating molecule have heat





Contd.

In 2004, US EPA tightened up nitrogen oxide emission standards by diesels by approximately 94%. Most automakers saw these new standards as a reason not to make diesels. VW sees new standards as a 'challenge'.

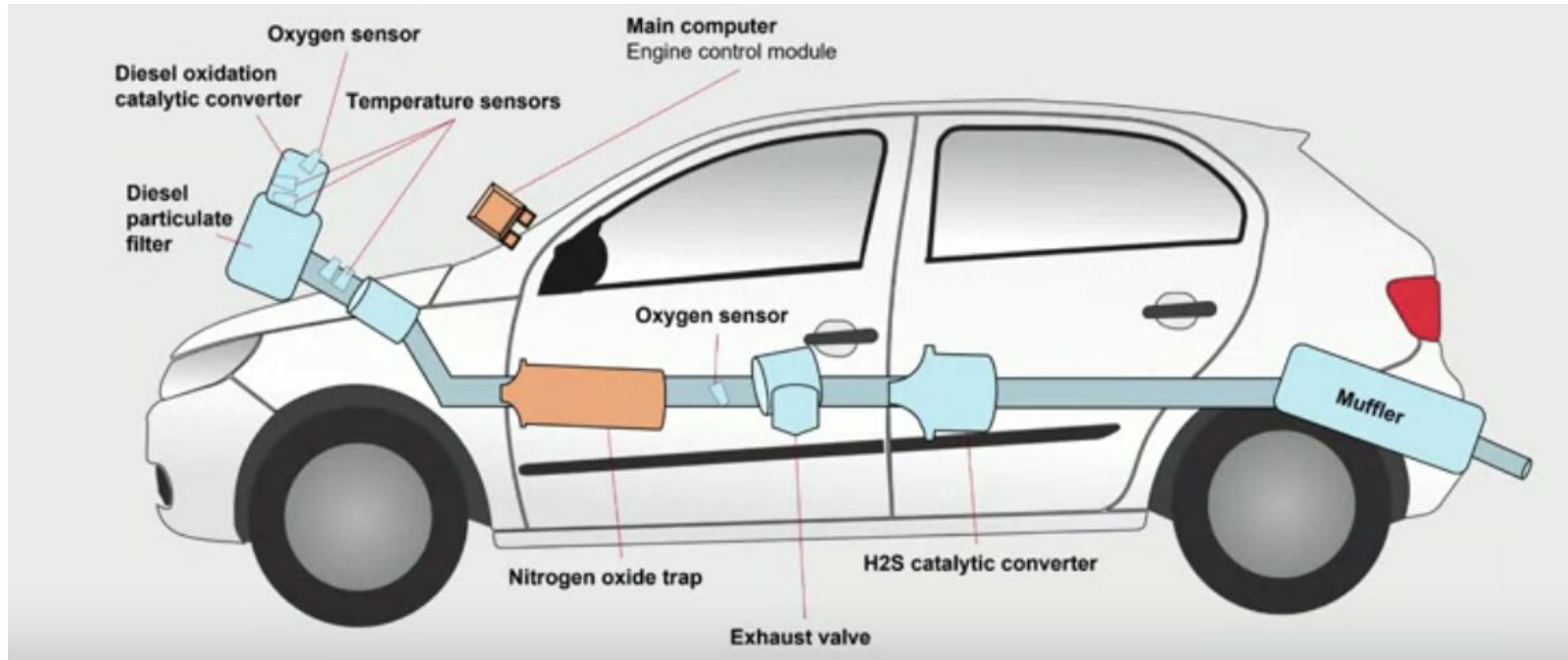
In 2007, the EU introduces new rules that prohibit car makers from using defeat devices.



The Opportunity

- Between 2004 and 2006 most global automobile manufacturers see that these are too strict and chose not to manufacture diesel engines
- 2006 - Alleged presentation to VW Execs re/overcoming EPA regulations.
- 2006 - VW invests in fuel efficient diesel engines as a way to increase the market share in US
- November 2006 - Execs decide to install defeat devices to detect when emissions testing is taking place and trigger pollution controls.

How Emission Control Works





Emission Software

Version 1 is the normal diesel that emits high amounts of CO₂ or nitrogen oxide

Version 2 - Vehicle realizes that it's in the lab and runs special software that gives completely different CO₂ and NO_x to regulators.

How does a Vehicle know that it's in a lab or not ?



Reflection

The engineers at Volkswagen developed two versions of software. V1 which runs 'normally' and emits high amounts of CO2 and NOx. Along with V2 which runs a special software that reads lower CO2 and NOx levels. Ethically speaking:

- If the V2 software was already developed, why did Volkswagen just implement that alone?



The Events

- 2011: the European Commission's Joint Research Center finds levels of harmful NOX coming out of Volkswagen's to exceed regulated EU levels by 14 times in different car models
- 2013: actually, a small team of students and researchers from a university in West Virginia, win the \$70,000 grant from the International Council on Clean Transportation to compare on-road versus in-lab results
- Road tests revealed, two VW emit far more NOX (35x) than allowed
- Volkswagen vehicles found to have 'defeat device'
- Requires major software effort to determine if the vehicle is in the lab vs on the road
 - Measure in-lab status
 - Could NOT have been accompanied by one person
 - Would need sufficient validation testing

Contd.

- May 2014, 2009-2015, two liter diesel engines - defeat devices
- 500K vehicles in the US and 11 million vehicles worldwide
- Suspicious test results sent to the CARB, California Air Resources Board
- VW recalled vehicles and re-flashed vehicle software, but did not remove the defeat device
- Further Test in May 2015 showed that the vehicle is still exceeded the limits by a lot
- September 19, 2015: Public learns of the fraud
- Non-compliance with the US Clean Air Act
- Caught having installed these defeat devices on the emission system
- November of 2015: VW implies that the gas vehicles were also affected
- Additional 800K vehicles
- Audi and Porsche 3.0 liter diesel also found to have the defeat software involved
- Nov 2016: Volkswagen finally admits broad knowledge



Reflection

Defeat Devices require an enormous amount of engineering effort. This was not a feat that could have been carried out with just one person.

- Why would a highly reputable company want its employees to do ***unethical work***?



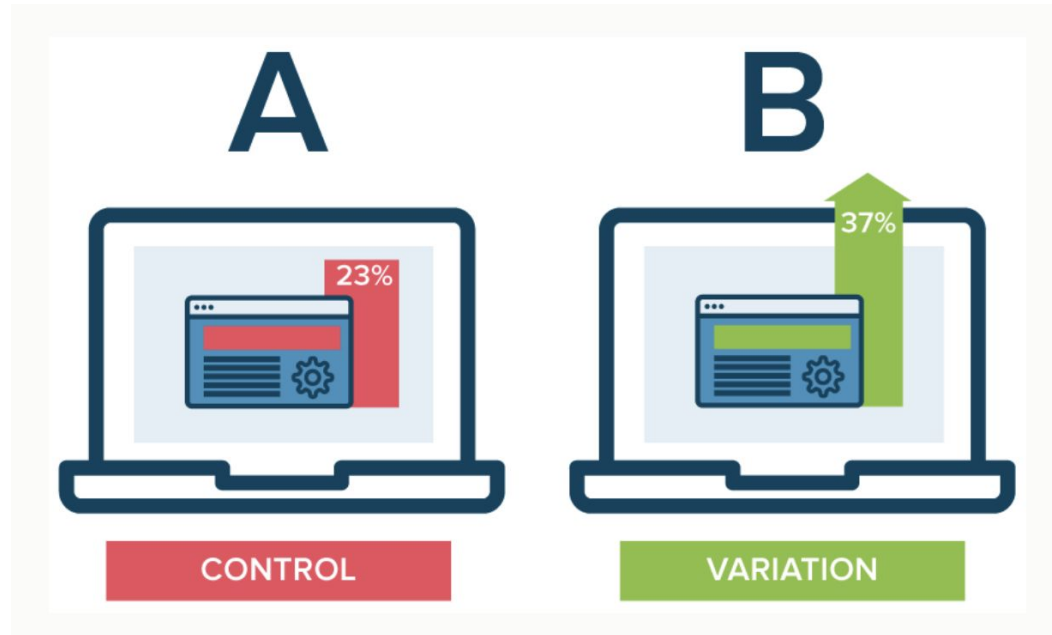
Aftermath - Organizational & Consumer Confidence

- Resignation of the CEO (Martin WinterKorn)
- Eight current and former executives have been charged with crime. Six of whom have been arrested
- 37 people are expected to take part in the fraud
- VW market share decreased by $\frac{1}{3}$
- Value of all VW vehicles decreased by 15%
- Pre Scandal - VW was world 18th most valuable brand
 - Currently it's 29th

Lessons Learned

- Process Improvement
- Obtain multiple sources of data in lab vs on road
- Create legal mechanism to enforce compliance
- Building test/checks into software development, particularly when regulation based
- Conflicting incentives may lead to suboptimal (illegal) solutions

Morality of A/B Testing





Further Reading

- ACM/IEEE-CS: Software Engineering Code of Ethics
 - <https://www.computer.org/web/education/code-ofethics>
- Vallor and Narayanan. An Introduction to Software Engineering Ethics
 - <https://www.scu.edu/media/ethicscenter/technology-ethics/Students.pdf>



Stakeholder Engagement

Stakeholders are specific groups of people (ex: not the general public), each with different desires and needs from the organization. A stakeholder is anyone who has a stake in your organization, either through interest, influence or both. Stakeholders can range from shareholders, to staff, board members, volunteers, funders, government, customers and beyond.

Stakeholder engagement is a process that organizations can follow in order to listen to, collaborate with, or inform (or a combination of all three) their existing stakeholders.

Identifying, mapping and prioritizing stakeholders to determine the best tactics for effective communication while making the best use of available resources.

Stakeholder Mapping

- **Low Interest & Low Influence**

For stakeholders with low interest and low influence, one-way communication of essential information will likely be sufficient in most instances.

- **Low Interest & High Influence**

This group is more *influential* than the low-low grouping. Although they still have a low interest in regular communication from your organization, it's important to monitor this group regularly, keep them informed of critical information, and anticipate any interests or needs they may have.

- **High Interest & Low Influence**

This group is more *interested* than the low-low grouping. Although they may not be as influential as other stakeholders, you may wish to have regular, robust communication and consider two-way communication with this highly interested stakeholder group.

- ***High Interest & High Influence (these are your KEY stakeholders)**

This group is both *interested* and *influential* and will require more resources to engage with effectively. This is your priority grouping of stakeholders who will require regular, robust, two-way communication tactics to keep this group satisfied

STAKEHOLDER MAPPING

Stakeholder mapping is an exercise to help you determine who your stakeholders are and how much engagement, communication or consideration they need.

By mapping and prioritizing your stakeholders, you can focus your attention in the most impactful way.

STAKEHOLDER INTEREST

HIGH INTEREST
+
LOW INFLUENCE

- Inform completely
- Monitor Closely

LOW INTEREST
+
LOW INFLUENCE

- Essential information
- Minimal contact

HIGH INTEREST
+
HIGH INFLUENCE

- Regularly engage
- Keep satisfied

LOW INTEREST
+
HIGH INFLUENCE

- Monitor regularly
- Anticipate needs

STAKEHOLDER INFLUENCE



Working with Senior Management

Managing up

- Understanding your manager/supervisor
- Understanding the goals
- Open communication
- Good Relation = More Effective



Have Backbone

- Work to understand managerial direction and directives first
- Don't be afraid to ask questions or pushback
- Critical thought is valued
- Sometimes you have to work on things even when you might disagree
 - Changes might be part of broader strategy
 - Recognize you could be wrong
 - Don't complain and just make it happen

How to Deliver Bad News

- Things always go wrong
- Be honest and don't blame it to others
- Talk about facts and come with solutions
 - Presenting plan to fix immediate problem
- Understand ROOT CAUSE issue
- How to avoid it in the future or actions taken to avoid reoccurrence



Meeting Etiquette - Social Skills 101

- If you don't know the answer, say, "I don't know" and don't make something up.
- Don't make an estimate based on your best guess. Just say, "I don't know," but say that you will find out
- Don't interrupt others
- Speak clearly and confidently
- Be Prepared
- Be on time and leave on time
- Don't call unnecessary meetings



Overcoming Introversion

- Extroverts tends to dominate discussion
- Find openings to talk without interrupting is hard
- Find an ally such as your manager or peer
- Speak less but when speaking say something insightful
- Listening more than speaking
- Structure the meeting , so that everyone gets time to speak
- Follow-up in written email



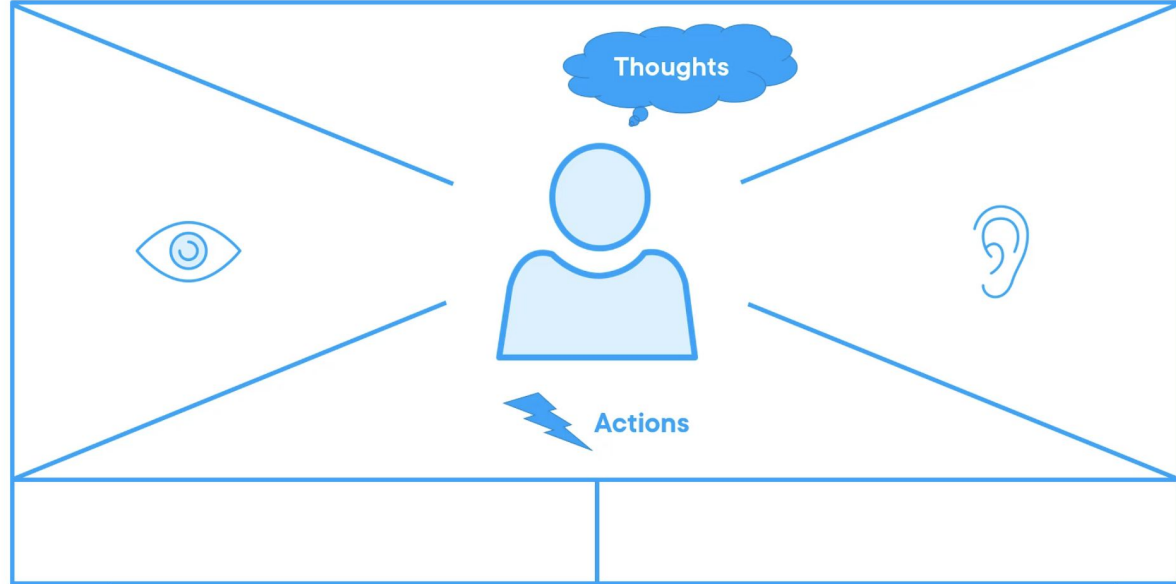
Presenting to Stakeholders

- Be prepared:
- Anticipate questions and have the answers and data ready
- Practice and run it by your manager or peers.
- Be succinct
- Don't take things personally
- Don't make up stuff, if something is unknown just say "I don't know and will get back to you right after the meeting"

Empathy

Understanding emotions of stakeholder and putting people first

Empathy Maps





Conflict Resolution

Identify root causes

**Find mutually
acceptable solution**

**Communicate openly
and honestly**



Final Word

- Thank you for giving us the privilege to teach you this semester.
- Please consider filling in the course evaluation forms.
- Stay connected. Here's my linkedin-page
 - <https://www.linkedin.com/in/agrawal-pankaj/>



Thank You